Circular motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate inward net force to a change in velocity direction.

SUCCESS CRITERIA - I can...

1. Draw tangent velocity and inward acceleration.
2. Use v^2/r or a supplied acceleration.
3. Identify the real interaction providing centripetal force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Two balls launch horizontally from the same height at 3 and 15 m/s. Ignore air resistance. Compare fall times and horizontal ranges.

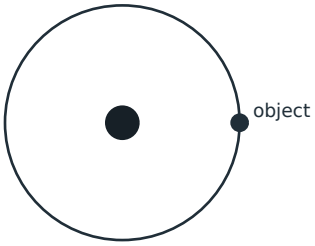
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $a_c = v^2/r$; $F_{net} = mv^2/r$.
c means centripetal: toward the center. This is the net inward force, not an extra force.

Worked reasoning: A 2 kg object moves at 3 m/s in a circle of radius 3 m. Inward acceleration is $v^2/r = 3\ m/s^2$; net force is 6 N inward.

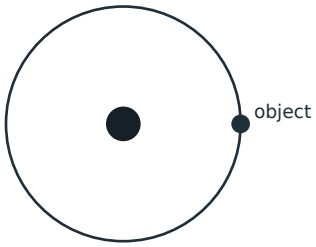
Your turn: Draw tangent velocity and inward net force at two different points.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Add velocity and acceleration arrows to the marked point on the circle. Name one interaction that could provide the inward force.

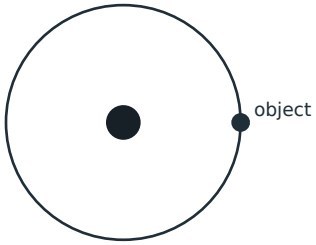


The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

04 Apply the physics

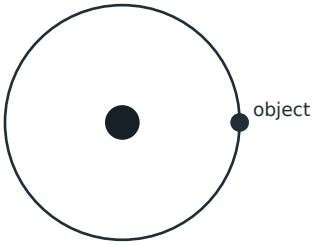
A 2 kg model car follows a circular track. Its inward acceleration is 3 m/s^2. Draw and label the inward net force and find its magnitude. How would that force change if inward acceleration doubled?



Marzano check - circle: ① ② ③

05 Explain your reasoning

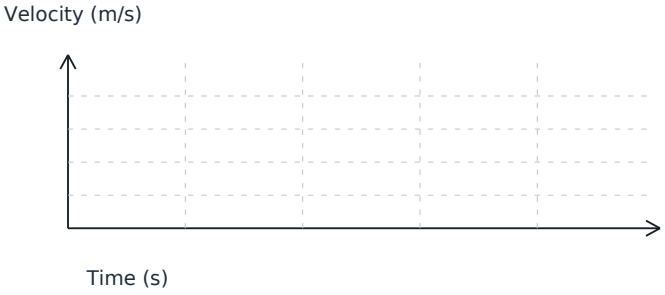
A string breaks while a ball moves in a circle. Neglect other forces over the next instant. Sketch its initial path after release.



Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

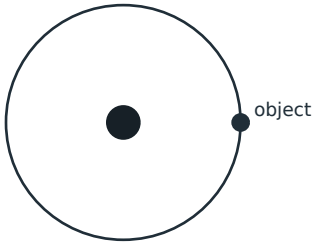
A 8 kg cart starts from rest under 80 N net force right for 7 s. Find acceleration and final velocity. State one assumption.
Label the given and predicted quantities on the model.



Marzano check - circle: 1 2 3

07 Exit check

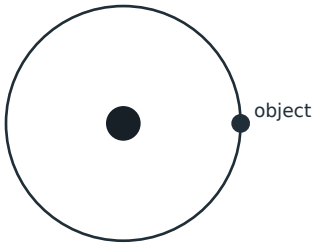
A car turns at steady speed. Can net force be zero? Explain using velocity. Label the radius and show velocity and inward acceleration on the circle.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2. At the same radius and mass, compare forces for speeds 3 and 6 m/s. Why is the answer not just twice as much? Label the radius and show velocity and inward acceleration on the circle.



Marzano check - circle: 1 2 3